

To install Apache Airflow on Ubuntu, you can follow these steps:

Step 1: Install Dependencies

1. Update Ubuntu packages:

```
sudo apt update
```

```
sudo apt upgrade
```

2 Install Python and pip: Ensure Python 3 and pip are installed:

```
sudo apt install python3 python3-pip
```

Step 2: Set Up a Virtual Environment (Optional but Recommended)

To avoid conflicts with other Python packages, it's a good idea to use a virtual environment.

Install virtualenv:

```
sudo pip3 install virtualenv
```

Create and activate a virtual environment:

```
virtualenv airflow_venv
```

```
source airflow_venv/bin/activate
```

Step 3: Install Apache Airflow

Set the Airflow version and install Airflow: Export the environment variables to specify Airflow version and extras:

```
export AIRFLOW_VERSION=2.7.0
```

```
export PYTHON_VERSION="$(python3 --version | cut -d " " -f 2 | cut -d "." -f 1-2)"
```

```
export CONSTRAINT_URL="https://raw.githubusercontent.com/apache/airflow/constraints-${AIRFLOW_VERSION}/constraints-${PYTHON_VERSION}.txt"
```

Install Apache Airflow with the desired extras: You can install extras like postgres, mysql, celery, aws, etc., based on your needs. Here, we'll install Airflow with mysql support:

```
pip install "apache-airflow[mysql]"==$AIRFLOW_VERSION --constraint "$CONSTRAINT_URL"
```

Step 4: Initialize Airflow Database:

Set the Airflow home directory:

```
export AIRFLOW_HOME=~/.airflow
```

Initialize the Airflow metadata database:

```
airflow db init
```

Step 5: Create an Admin User:

Create an admin user for the Airflow web interface:

```
airflow users create --username admin --password admin --firstname Admin --lastname User  
--role Admin --email admin@example.com
```

Step 6: Start Airflow Services

1. Start the Airflow webserver (runs on port 8080):

```
airflow webserver --port 8080
```

2. Start the Airflow scheduler: In a new terminal (or tab), run:

```
airflow scheduler
```

You can now access the Airflow web UI by going to `http://localhost:8080` in your browser.